Eighth Grade Tile Task

Standard 8.NS.1

8

1. a) Tenths digit of 0.444... 4

b) Tenths digit of 0.999... 9

c) Square root of 0 0

2. a) Tenths digit of 0.222... 2

b) Square root of 1 1

c) Square root of 36 6

3. a) Square root of 9 3

b) Square root of 49 7

c) Square root of 25 5

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.NS.2

8

1. a) Whole part of $\sqrt{0}$ **0**

b) Whole part of $\sqrt{81}$ **9**

c) Whole part of $\sqrt{47}$ **6**

2. a) Whole part of $\sqrt{15}$ **3**

b) Whole part of $\sqrt{33}$ **5**

c) Whole part of $\sqrt{7}$ **2**

3. a) Whole part of $\sqrt{2}$ **1**

b) Whole part of $\sqrt{57}$ **7**

c) Whole part of $\sqrt{79}$ **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.1

8

1. **a)** x^2 × x^4 = x^?: build the exponent — 6

   b) x^5 × x^4 = x^?: build the exponent — 9

   c) x^0 × x^7 = x^?: build the exponent — 7

2. **a)** x^1 × x^0 = x^?: build the exponent — 1

   b) x^2 × x^0 = x^?: build the exponent — 2

   c) x^0 × x^0 = x^?: build the exponent — 0

3. **a)** x^1 × x^3 = x^?: build the exponent — 4

   b) x^5 × x^0 = x^?: build the exponent — 5

   c) x^7 × x^1 = x^?: build the exponent — 8

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: EIGHTH GRADE

Eighth Grade Tile Task

Standard 8.EE.2

8

1. a) Square root of 25 — 5

b) Square root of 16 — 4

c) Square root of 1 — 1

2. a) Square root of 9 — 3

b) Square root of 49 — 7

c) Cube root of 8 — 2

3. a) Square root of 81 — 9

b) Square root of 36 — 6

c) Cube root of 0 — 0

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.3

8

1. **a)** Exponent in 1 × 10^k — 9
   b) Exponent in 2 × 10^k — 2
   c) Exponent in 9 × 10^k — 3

2. **a)** Exponent in 5 × 10^k — 8
   b) Exponent in 5 × 10^k — 5
   c) Exponent in 4 × 10^k — 1

3. **a)** Exponent in 5 × 10^k — 7
   b) Exponent in 8 × 10^k — 0
   c) Exponent in 5 × 10^k — 6

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.4

8

1. a) Exponent in 6 × 10^k for the number 60000000 — 7

b) Exponent in 3 × 10^k for the number 300000 — 5

c) Exponent in 9 × 10^k for the number 9000 — 3

2. a) Exponent in 7 × 10^k for the number 700000000 — 8

b) Exponent in 6 × 10^k for the number 600 — 2

c) Exponent in 6 × 10^k for the number 60000 — 4

3. a) Exponent in 8 × 10^k for the number 8000000 — 6

b) Exponent in 7 × 10^k for the number 7000000000 — 9

c) Exponent in 7 × 10^k for the number 70 — 1

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.5

8

1. a) Slope of y = 2x + 2 — 2

b) Slope of y = 8x + 3 — 8

c) Slope of y = 5x + 1 — 5

2. a) Slope of y = 9x + 1 — 9

b) Slope of y = 6x + 3 — 6

c) Slope of y = 7x + 3 — 7

3. a) Slope of y = 1x + 3 — 1

b) Slope of y = 3x + 2 — 3

c) Slope of y = 0x + 1 — 0

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.6

8

1. a) y-intercept of y = 3x + 2 — 2

b) y-intercept of y = 0x + 9 — 9

c) y-intercept of y = 3x + 0 — 0

2. a) y-intercept of y = 1x + 8 — 8

b) y-intercept of y = 1x + 5 — 5

c) y-intercept of y = 2x + 7 — 7

3. a) y-intercept of y = 1x + 1 — 1

b) y-intercept of y = 4x + 3 — 3

c) y-intercept of y = 3x + 4 — 4

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.7

8

1. a) Solve 5x + 8 = 28: x = ? **4**

b) Solve 2x + 5 = 11: x = ? **3**

c) Solve 4x + 6 = 14: x = ? **2**

2. a) Solve 9x + 9 = 90: x = ? **9**

b) Solve 4x + 0 = 24: x = ? **6**

c) Solve 4x + 9 = 29: x = ? **5**

3. a) Solve 4x + 8 = 8: x = ? **0**

b) Solve 6x + 3 = 51: x = ? **8**

c) Solve 4x + 7 = 35: x = ? **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.8

8

1. a) Solve x + y = 14, x - y = 0: x = ? **7**

b) Solve x + y = 1, x - y = 1: x = ? **1**

c) Solve x + y = 5, x - y = 1: x = ? **3**

2. a) Solve x + y = 6, x - y = 4: x = ? **5**

b) Solve x + y = 0, x - y = 0: x = ? **0**

c) Solve x + y = 15, x - y = 3: x = ? **9**

3. a) Solve x + y = 7, x - y = 1: x = ? **4**

b) Solve x + y = 11, x - y = 1: x = ? **6**

c) Solve x + y = 3, x - y = 1: x = ? **2**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.1

8

1. a) f(x) = 3x + 2. Find f(0). **2**

b) f(x) = 1x + 2. Find f(1). **3**

c) f(x) = 1x + 1. Find f(0). **1**

2. a) f(x) = 2x + 0. Find f(0). **0**

b) f(x) = 1x + 5. Find f(2). **7**

c) f(x) = 3x + 0. Find f(2). **6**

3. a) f(x) = 2x + 4. Find f(0). **4**

b) f(x) = 2x + 3. Find f(3). **9**

c) f(x) = 1x + 5. Find f(0). **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.2

8

1. a) Slope of function A: y = 6x — 6

b) Slope of function A: y = 4x — 4

c) Slope of function A: y = 3x — 3

2. a) Slope of function A: y = 8x — 8

b) Slope of function A: y = 2x — 2

c) Slope of function A: y = 0x — 0

3. a) Slope of function A: y = 7x — 7

b) Slope of function A: y = 1x — 1

c) Slope of function A: y = 9x — 9

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.3

8

1. a) $f(x) = x^2 + 5$. Find $f(1)$. 6

b) $f(x) = x^2 + 2$. Find $f(1)$. 3

c) $f(x) = x^2 + 7$. Find $f(0)$. 7

2. a) $f(x) = x^2 + 8$. Find $f(0)$. 8

b) $f(x) = x^2 + 4$. Find $f(0)$. 4

c) $f(x) = x^2 + 0$. Find $f(0)$. 0

3. a) $f(x) = x^2 + 1$. Find $f(1)$. 2

b) $f(x) = x^2 + 8$. Find $f(1)$. 9

c) $f(x) = x^2 + 1$. Find $f(2)$. 5

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.4

8

1. **a)** Rate of change in y = 3x + 2 — 3

   b) Rate of change in y = 9x + 1 — 9

   c) Rate of change in y = 4x + 2 — 4

2. **a)** Rate of change in y = 1x + 1 — 1

   b) Rate of change in y = 7x + 1 — 7

   c) Rate of change in y = 6x + 1 — 6

3. **a)** Rate of change in y = 0x + 0 — 0

   b) Rate of change in y = 5x + 2 — 5

   c) Rate of change in y = 2x + 2 — 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.5

8

1. **a)** y-intercept of a line through (0, 0) — 0
   b) y-intercept of a line through (0, 5) — 5
   c) y-intercept of a line through (0, 2) — 2

2. **a)** y-intercept of a line through (0, 9) — 9
   b) y-intercept of a line through (0, 1) — 1
   c) y-intercept of a line through (0, 3) — 3

3. **a)** y-intercept of a line through (0, 4) — 4
   b) y-intercept of a line through (0, 8) — 8
   c) y-intercept of a line through (0, 7) — 7

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.1

8

1. a) Reflect (3, 1) across the x-axis: new y is? **1**

b) Reflect (3, 3) across the x-axis: new y is? **3**

c) Reflect (3, 7) across the x-axis: new y is? **7**

2. a) Reflect (3, 6) across the x-axis: new y is? **6**

b) Reflect (3, 0) across the x-axis: new y is? **0**

c) Reflect (3, 5) across the x-axis: new y is? **5**

3. a) Reflect (3, 8) across the x-axis: new y is? **8**

b) Reflect (3, 2) across the x-axis: new y is? **2**

c) Reflect (3, 9) across the x-axis: new y is? **9**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.2

8

1. **a)** Sides on a pentagon? **5**

   b) Sides on a square? **4**

   c) Sides on a heptagon? **7**

2. **a)** Sides on an octagon? **8**

   b) Sides on a hexagon? **6**

   c) Intersection points of two parallel lines? **0**

3. **a)** Sides on a nonagon? **9**

   b) Intersection points of two distinct lines? **1**

   c) Lines needed to form a triangle? **3**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.3

8

1. **a)** Dilate a 3 cm segment by 3: new length? 9

   b) Dilate a 1 cm segment by 1: new length? 1

   c) Dilate a 2 cm segment by 1: new length? 2

2. **a)** Dilate a 7 cm segment by 1: new length? 7

   b) Dilate a 2 cm segment by 2: new length? 4

   c) Dilate a 1 cm segment by 8: new length? 8

3. **a)** Dilate a 2 cm segment by 3: new length? 6

   b) Dilate a 5 cm segment by 1: new length? 5

   c) Dilate a 3 cm segment by 1: new length? 3

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.4

8

1. a) Scale factor from 2 to 16: build the factor — 8

b) Scale factor from 2 to 6: build the factor — 3

c) Scale factor from 2 to 10: build the factor — 5

2. a) Scale factor from 2 to 18: build the factor — 9

b) Scale factor from 2 to 4: build the factor — 2

c) Scale factor from 2 to 12: build the factor — 6

3. a) Scale factor from 2 to 14: build the factor — 7

b) Scale factor from 2 to 2: build the factor — 1

c) Scale factor from 2 to 8: build the factor — 4

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.5

8

1. **a)** Supplementary angle of 177° 3

   b) Supplementary angle of 178° 2

   c) Supplementary angle of 179° 1

2. **a)** Supplementary angle of 171° 9

   b) Supplementary angle of 180° 0

   c) Supplementary angle of 176° 4

3. **a)** Supplementary angle of 172° 8

   b) Supplementary angle of 175° 5

   c) Complementary angle of 83° 7

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.6

8

1. a) Value of $1^2 + 1^2$? **2**

b) Value of $3^2 - 2^2$? **5**

c) Value of $4^2 - 3^2$? **7**

2. a) Value of $0^2 + 0^2$? **0**

b) Value of $2^2 - 1^2$? **3**

c) Value of $2^2 + 2^2$? **8**

3. a) Value of $2^2 + 0^2$? **4**

b) Value of $3^2 + 0^2$? **9**

c) Value of $1^2 + 1^2 + 2^2$? **6**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.7

8

1. a) Area of a right triangle with legs 7 and 2? **7**

b) Area of a 3-4-5 right triangle? **6**

c) Area of a right triangle with legs 3 and 6? **9**

2. a) Area of a right triangle with legs 2 and 2? **2**

b) Area of a square with side 1? **1**

c) Hypotenuse 10, leg 6: other leg? **8**

3. a) Hypotenuse 5, leg 4: other leg? **3**

b) Area of a right triangle with legs 5 and 2? **5**

c) Area of a square with side 2? **4**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.8

8

1. a) Dilate point at x = 1 by 5: new x? **5**

b) Dilate point at x = 1 by 2: new x? **2**

c) Dilate point at x = 4 by 1: new x? **4**

2. a) Dilate point at x = 3 by 1: new x? **3**

b) Dilate point at x = 1 by 1: new x? **1**

c) Dilate point at x = 7 by 1: new x? **7**

3. a) Dilate point at x = 1 by 8: new x? **8**

b) Dilate point at x = 9 by 1: new x? **9**

c) Dilate point at x = 3 by 2: new x? **6**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.9

8

1. a) Volume of a 1 by 7 by 1 prism — 7

b) Volume of a 2 by 4 by 1 prism — 8

c) Volume of a 1 by 5 by 1 prism — 5

2. a) Volume of a 1 by 2 by 1 prism — 2

b) Volume of a 3 by 1 by 1 prism — 3

c) Volume of a 4 by 1 by 1 prism — 4

3. a) Volume of a cube with side 1 — 1

b) Volume of a cylinder, r=1, height 2 ($\pi \approx 3$) — 6

c) Volume of a 9 by 1 by 1 prism — 9

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.SP.1

8

1. **a)** Scatter plot with 7 data points: how many dots? **7**
   b) Scatter plot with 2 data points: how many dots? **2**
   c) Scatter plot with 4 data points: how many dots? **4**

2. **a)** Scatter plot with 3 data points: how many dots? **3**
   b) Scatter plot with 8 data points: how many dots? **8**
   c) Scatter plot with 1 data points: how many dots? **1**

3. **a)** Scatter plot with 9 data points: how many dots? **9**
   b) Scatter plot with 5 data points: how many dots? **5**
   c) Scatter plot with 6 data points: how many dots? **6**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.SP.2

8

1. **a)** Best-fit line y = 8x: predicted y at x = 1 — **8**

   b) Best-fit line y = 2x: predicted y at x = 1 — **2**

   c) Best-fit line y = 6x: predicted y at x = 1 — **6**

2. **a)** Best-fit line y = 0x: predicted y at x = 1 — **0**

   b) Best-fit line y = 5x: predicted y at x = 1 — **5**

   c) Best-fit line y = 1x: predicted y at x = 1 — **1**

3. **a)** Best-fit line y = 9x: predicted y at x = 1 — **9**

   b) Best-fit line y = 3x: predicted y at x = 1 — **3**

   c) Best-fit line y = 7x: predicted y at x = 1 — **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.SP.3

8

1. a) Slope in y = 2x + 0 — 2

b) Slope in y = 4x + 1 — 4

c) Slope in y = 1x + 1 — 1

2. a) Slope in y = 3x + 1 — 3

b) Slope in y = 0x + 1 — 0

c) Slope in y = 8x + 0 — 8

3. a) Slope in y = 5x + 0 — 5

b) Slope in y = 6x + 0 — 6

c) Slope in y = 9x + 0 — 9

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.SP.4

8

1. a) Two-way table: 1 boys and 0 girls. Total? — 1

b) Two-way table: 3 boys and 3 girls. Total? — 6

c) Two-way table: 0 boys and 8 girls. Total? — 8

2. a) Two-way table: 2 boys and 1 girls. Total? — 3

b) Two-way table: 0 boys and 2 girls. Total? — 2

c) Two-way table: 0 boys and 5 girls. Total? — 5

3. a) Two-way table: 4 boys and 0 girls. Total? — 4

b) Two-way table: 1 boys and 8 girls. Total? — 9

c) Two-way table: 1 boys and 6 girls. Total? — 7

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9